

CAPSTONE PROJECT REGISTRATION**1. General Information**

Item	Details
Project Title	Smart Elderly Care AI - A Multimodal AI System for Proactive Monitoring, Forecasting, and Care of Elderly Aged 60 and Above.
Project Type	☐ Industry-based ☐ Research-based ☒ Startup/Product ☐ Others: ____
Capstone Course	☒ CMU-SE 450 ☐ CMU-SE 451 ☐ CMU-IS 450 ☐ CMU-IS 451 ☐ CMU-CS 450 ☐ CMU-CS 451
Semester / Year	2026-2027
Proposed by	☒ Student Team ☐ Faculty ☐ Industry Partner

2. Team Information

No.	Full Name	Student ID	Email	Role (Leader/Dev/QA/BA/...)
1	Hieu,Duong Van	29211155327	hdvan884@gmail.com	Leader, Scrum Master, QA, Developer, Tester
2	Duy,Tran Van	29211155992	tvduy2807@gmail.com	AI Developer, Backend Developer, Frontend Developer, Tester
3	My,Phan Nguyen Giang	29201155816	1802giangmy@gmail.com	BA, UX/UI Designer, Frontend Developer, Backend Developer, Tester, Document
4	Nghia,Nguyen Huu	29219022025	ngghia12072003@gmail.com	AI Developer, Backend Developer, Frontend Developer, Tester
5	Vinh,Ngo Le	29211157943	ngolevinh233@gmail.com	Backend Developer, Frontend Developer, Tester

3. Supervisor / Mentor

Role	Name	Organization	Email
Faculty Mentor	PhD Dieu,Huynh Ba	DTU	dieuhb@gmail.com
Industry Mentor (if any)			

4. Problem Statement

- Background of the problem:

The global demographic shift towards an aging population has severely escalated the need for effective, continuous elderly care. Individuals in the 60-89 age demographic are highly vulnerable to sudden domestic accidents, most notably falls, as well as silent, gradual declines in cardiovascular, respiratory, and cognitive functions. Traditional caregiving is highly resource-intensive and prone to human error, creating a pressing demand for automated, technology-driven systems that can transition elderly care from a reactive emergency response model to a proactive health monitoring approach.

- Current limitations / gaps:

Existing monitoring solutions in the market possess critical limitations. First, they are fundamentally reactive and suffer from high false-positive rates, typically triggering alerts only *after* a severe incident has occurred. Second, these systems rely almost exclusively on standard visual (RGB) cameras, creating severe privacy violations. Consequently, they cannot be deployed in sensitive, high-risk environments like bathrooms or bedrooms—where the majority of fatal falls occur. Furthermore, current systems lack the capability to passively measure continuous vital signs (such as respiration rate or body temperature) without forcing the elderly to wear intrusive, specialized medical devices.

- Target users / stakeholders:
 - **Primary Target Users:** Elderly individuals (aged 60-89) living independently in smart homes or residing in assisted living facilities/nursing homes.
 - **Secondary Users:** Family members and legal guardians who require real-time health updates and emergency notifications.
 - **Stakeholders:** Nursing home administrators, healthcare professionals (doctors/nurses), and emergency medical responders who utilize the system's generated data (Digital Twin) to optimize care delivery and resource allocation.

5. Project Objectives

- **Objective 1:** Cross-Platform Software Development (Month 1) To build and deploy a secure Cloud Backend, Web Dashboard, and Mobile App featuring a 3-tier escalation alert workflow, ensuring that 100% of simulated high-risk notifications are successfully delivered to designated caregivers within 3 seconds, by the end of Month 1.
 - **Specific:** Develop the Web Dashboard, Mobile App, and the Cloud API interfaces using mock data for initial testing.

- **Measurable:** 100% delivery rate of simulated high-risk alerts within a 3-second transmission window.
- **Achievable:** Utilizes standard web and mobile frameworks (e.g., React, Flask) within a focused one-month sprint.
- **Relevant:** Forms the critical user-facing layer of the monitoring system for caregivers and administrators.
- **Time-bound:** Strictly completed by the end of Month 1.
- **Objective 2:** Comprehensive Engineering Documentation (Month 2) To draft and finalize all mandatory project documentation, specifically the Software Requirements Specification (SRS), System Architecture Design (SAD), and API specifications, achieving compliance with required software engineering standards by the end of Month 2.
 - **Specific:** Create detailed SRS, SAD, and complete system requirement documents based on the Month 1 software build.
 - **Measurable:** 100% completion of documentation deliverables required for the midterm evaluation.
 - **Achievable:** Capitalizes on the established software architecture designed in the previous month.
 - **Relevant:** Crucial for academic evaluation, project sustainability, and preparing a clear blueprint for hardware integration.
 - **Time-bound:** Strictly completed by the end of Month 2.
- **Objective 3:** Hardware Integration, AI Testing & Final Delivery (Month 3 & 4) To assemble the Edge AI hardware prototype (Raspberry Pi, NoIR, Thermal, Audio), integrate the Multimodal AI models, and achieve a detection accuracy of >90% with a false-positive rate of <5% during physical testing in Month 3, followed by successful end-to-end system validation and final submission by the end of Month 4.
 - **Specific:** Assemble physical hardware, test embedded ML models, and finalize the complete Edge-to-Cloud system connection.
 - **Measurable:** System maintains >90% detection accuracy, <5% false-positive rate, and complete submission of all final deliverables.

- **Achievable:** Systematically combines Month 1 software and Month 3 hardware into a finalized, tested product.
- **Relevant:** Fulfills the core requirement of delivering a functional, real-world Multimodal AI healthcare system.
- **Time-bound:** Hardware assembled and tested by Month 3; final project submission and demonstration by the end of Month 4.

6. Proposed Solution

System overview

The proposed solution is a privacy-preserving Multimodal AI Healthcare System that leverages Edge Computing (Orange Pi 5 (Rockchip RK3588S NPU)) to proactively monitor elderly individuals. By simultaneously fusing data from four distinct sources—a NoIR Camera for pose estimation, a Thermal Sensor (AMG8833) for passive temperature estimation, a 4-Mic Array for audio event detection, and a BLE wristband for vital signs monitoring (heart rate, SpO2, and impact acceleration)—the system analyzes health anomalies locally. To ensure absolute privacy, no raw video or audio is transmitted; only encrypted metadata and Risk Scores are pushed to a centralized Cloud Backend, which instantly synchronizes with cross-platform applications (Web and Mobile) for real-time caregiver monitoring.

Key features

- **Multimodal Sensor Fusion:** Synchronized, passive analysis of body posture (fall detection), thermal variations (vital sign estimation), and acoustic anomalies (cries for help, loud impacts).
- **Edge AI Privacy Architecture:** 100% of raw media processing occurs locally on the edge device, entirely eliminating the privacy risks associated with traditional cloud-streaming cameras.
- **3-Tier Escalation Alert Workflow:** An automated response mechanism that classifies events into Notification, Warning, and Emergency levels, automatically escalating alerts to higher authorities (e.g., nursing home administration) if left unacknowledged by primary caregivers.
- **Cross-Platform Health Digital Twin:** A React-based Web Dashboard and Mobile App that visualize real-time alerts and historical health trends (time-series data) to assist medical professionals in making data-driven decisions.

Expected outputs

- **01 Functional Edge AI Hardware Prototype:** Fully assembled with Raspberry Pi, NoIR Camera, Thermal Sensor (AMG8833), and a 4-Mic Array.
- **01 Deployed Cloud Backend & Database:** Robust RESTful APIs (Python Flask) and a SQL database for secure metadata storage.
- **01 Web Dashboard & 01 Mobile Application:** Cross-platform monitoring interfaces featuring the escalation workflow and real-time data visualization.
- **Comprehensive Engineering Documentation:** Software Requirements Specification (SRS), System Architecture Design (SAD), API specifications, and empirical testing reports.

7. Complex Engineering Problem Justification

Criteria	Yes/No	Explanation
Involving wide-ranging or conflicting technical issues. (e.g., security vs performance)	Yes	The project demands balancing heavy computational AI workloads (processing continuous RGB, thermal, and audio feeds simultaneously) against the severe hardware constraints of an edge device (Orange Pi 5 (Rockchip RK3588S NPU)). Furthermore, it resolves the conflict between comprehensive monitoring and strict data privacy by processing all raw media locally, ensuring only metadata reaches the cloud.
No obvious existing solution	No	(Leave blank)
Involving diverse groups of stakeholders.	Yes	The system architecture must concurrently satisfy

Criteria	Yes/No	Explanation
		the passive, non-intrusive monitoring needs of elderly patients , provide real-time mobile alerts to family members , and supply actionable health data dashboards for medical professionals and nursing home administrators .
Multi-technology integration (AI, IoT, Big Data, etc.)	Yes	The system intricately integrates IoT hardware (Raspberry Pi, NoIR, Thermal, and Audio sensors), embedded Edge AI (machine learning models for pose and acoustic event detection), and full-stack software development (Cloud REST APIs, React Web Dashboard, and Mobile App) into a unified pipeline.
Significant societal/industrial impact	Yes	Addressing the critical healthcare crisis of an aging demographic, this proactive system prevents fatal delays in emergency responses for domestic accidents (e.g., falls). Industrially, it significantly optimizes resource allocation and reduces the severe workload and burnout experienced by professional caregivers.

Criteria	Yes/No	Explanation
Addressing problems not encompassed by current standards and codes.	No	(Leave blank)
Involving multiple disciplines.	No	(Leave blank)

8. System Constraints & Impact Analysis

8.1 Engineering Constraints

- ☒ Performance
- ☒ Security & Privacy
- ☐ Scalability
- ☒ Cost
- ☐ Maintainability
- ☐ Economic
- ☒ Ethical
- ☐ Public health, safety, and welfare
- ☒ Social and Global
- ☐ Cultural
- ☐ Sustainability
- ☐ Other: please specify.....

Explain how system addresses them:

- **Performance:** Running simultaneous Multimodal AI tasks (video, thermal, audio) on a resource-constrained edge device (Orange Pi 5 (Rockchip RK3588S NPU)) presents significant performance challenges. The system is designed to maintain emergency alert activation latency under 3 seconds, while MQTT alert packet transmission latency from the Edge Hub to the server and mobile application remains under 500 ms.
- **Security & Privacy:** Traditional cameras violate privacy in sensitive areas (bathrooms/bedrooms). This constraint is resolved via the Edge AI architecture; 100% of raw image, thermal, and audio data is processed and instantly discarded locally on the edge device. Only encrypted text-based metadata and Risk Scores are transmitted to the Cloud backend.
- **Cost:** Developing a medical-grade monitoring system is typically expensive. The system addresses this by using affordable, commercially available microcomputers and sensors

(Orange Pi 5 (Rockchip RK3588S NPU), AMG8833 Thermal Sensor, NoIR Camera), keeping the prototype cost highly accessible for mass deployment.

8.2 Global & Societal Impact

- **Cultural / Social impact:** The system promotes "aging in place," allowing the elderly to live independently with dignity and peace of mind, without feeling constantly watched by invasive traditional cameras. It also significantly reduces the psychological stress and anxiety experienced by family members.
- **Environmental (Green IT):** By utilizing low-power, ARM-based edge computing devices (Raspberry Pi) and only transmitting small packets of metadata (kilobytes) rather than continuous high-definition raw video streams (gigabytes), the system drastically reduces cloud server processing loads and overall energy consumption.
- **Economic impact:** The solution reduces the extreme financial burden associated with 24/7 in-person nursing care. Furthermore, by proactively detecting falls and health anomalies early, it minimizes the severe medical costs associated with delayed emergency treatments.

8.3 Ethical Considerations

- **Data privacy:** Strict adherence to patient confidentiality. Caregivers and administrators only see a dashboard of health metrics and alerts (Digital Twin), ensuring the physical dignity of the elderly user is never compromised, even in high-risk zones like restrooms.
- **AI bias / fairness:** To prevent AI bias, the algorithms focus on universal physiological and physical traits (e.g., skeletal angles for pose estimation, environmental acoustic anomalies, and thermal signatures) rather than facial recognition or skin tone analysis. The models will be calibrated to recognize various body shapes and mobility aids (e.g., walkers, wheelchairs) to ensure fairness across all user demographics.
- **Intellectual property #NA:** Open-source libraries (e.g., OpenCV, Python Flask, React) will be strictly utilized under their respective MIT/Apache licenses, with proper attribution in the documentation.

9. Standard requirements

- ☒ Code standards (GNU Coding Standards, Oracle Java Code Conventions, ...)
- ☒ Design standards (Design Patterns, Object-Oriented Analysis and Design, ...)
- ☒ IEEE standards (1058, 1540, 830, 1016, 829, 1012, 1008)
- ☐ ISO/IEC/IEEE 12207:2017 (TCVN 10539:2014); ISO/IEC 25051:2006 (TCVN 10540:2014)
- ☒ Other standards related to specific topics.

10. Technical Stack & Architecture

Category	Proposed Technology
Frontend Web	React.js
Backend	Python (FastAPI) for Cloud RESTful APIs;
Database	PostgreSQL (for relational data: users, digital twins, historical logs) and Firebase/Socket.io (for real-time push notifications).
DevOps / CI-CD	Docker (for containerizing the backend services), GitHub Actions (for automated testing and continuous integration).
Tools (Jira, Git, etc.)	Git/GitHub (Version Control), Jira (Agile/Scrum Task Management), Postman (API Testing), Figma (UI/UX Design).
Edge Device	Orange Pi 5 with Rockchip RK3588S NPU
Computer Vision	YOLOv8-Pose, OpenCV
Audio AI	YAMNet / audio event classification
Thermal	AMG8833 thermal sensor
Wearable	BLE wristband for Heart Rate, SpO2, Impact Acceleration
Mobile App	React Native
Push Notification	Firebase Cloud Messaging (FCM)
Cache / Real-time	Redis, EMQX MQTT

10. Development Methodology

- ☒ Agile/Scrum
- ☐ V-Model
- ☐ Hybrid

Sprint plan (high-level):

The project will be executed in 7 Sprints, with each Sprint lasting approximately 2 weeks, aligning with the project timeline from 21/08/2026 to 06/12/2026:

- **Sprint 1 - 2 (Month 1): Software Prototype & Mock Data.** Focus on setting up the database, building Cloud APIs, and developing the UI/UX for the React Web Dashboard and Mobile App using mock sensor data. Deliverable: Functional software platforms.
- **Sprint 3 - 4 (Month 2): Engineering Documentation.** Translate the established architecture into formal documentation. Focus on drafting and finalizing the Software Requirements Specification (SRS) and System Architecture Design (SAD). Deliverable: Completed mid-term documentation.
- **Sprint 5 - 6 (Month 3): Hardware Assembly & Edge AI Integration.** Assemble the Orange Pi 5 with NoIR Camera, AMG8833 Thermal Sensor, Audio Sensor, and BLE wristband. Deploy and test the embedded AI models (Pose Estimation, Audio Classification) directly on the edge device. Deliverable: Functional hardware prototype.
- **Sprint 7 (Month 4): System Validation & Final Delivery.** Perform end-to-end integration testing (Edge-to-Cloud synchronization). Optimize latency, conduct user acceptance testing (UAT), and prepare the final presentation and physical demonstration. Deliverable: Final polished product and final reports.

11. Major Deliverables Plan

Phase	Deliverable
Proposal	Project Proposal
Design	SRS + SAD
Implementation	Prototype + Test
Final	Report + Demo

12. Risk Assessment

Risk	Impact	Mitigation
Edge Device Processing Overload	High: High latency (>2s) or dropped frames when running concurrent multimodal models (Pose, Thermal, Audio) on a resource-constrained Orange Pi 5 (Rockchip RK3588S NPU).	Utilize lightweight AI models (e.g., MediaPipe, TensorFlow Lite). Implement multi-threading and down-sample video/thermal input frame rates to optimize CPU/RAM usage.

Risk	Impact	Mitigation
Low Resolution of Thermal Sensor (AMG8833)	Medium: Difficulty in accurately extracting fine physiological data (e.g., respiration rate) from an 8x8 pixel matrix.	Apply bi-cubic image interpolation algorithms to upscale the thermal matrix. Rely on multi-modal cross-validation (verifying thermal anomalies with audio/pose data).
Hardware Delivery Delays / Component Failure	Medium: Significant delays in Month 3 hardware assembly and physical testing phases.	Develop Month 1 and Month 2 software architecture using simulated mock data and API endpoints. Order critical components immediately upon project approval.
Edge-to-Cloud Network Instability	High: Failure to deliver emergency alerts to the Web Dashboard and Mobile App within the critical 3-second window.	Implement local caching (SQLite) on the edge device to queue unsent data. Trigger a loud physical audio alarm directly from the Orange Pi 5 as a local failsafe if the Wi-Fi connection drops.

13. Expected Contributions

- Technical contribution** The project demonstrates the successful integration and optimization of concurrent multimodal machine learning pipelines (pose estimation, acoustic event classification, and thermal data processing) on a resource-constrained edge device (Orange Pi 5 (Rockchip RK3588S NPU)). Furthermore, it establishes a robust, low-latency Edge-to-Cloud data synchronization architecture, seamlessly connecting embedded Python processing with a scalable full-stack ecosystem (React Web Dashboard and Mobile App) via RESTful APIs.
- Innovation / novelty** The core innovation lies in shifting elderly care from a reactive, privacy-invasive model to a proactive, privacy-preserving paradigm. By utilizing Edge AI to process and immediately discard raw multimodal data (NoIR, Thermal, Audio) locally, the system completely protects user privacy while successfully identifying both sudden emergencies (falls) and gradual anomalies (vital sign changes). The implementation of a multi-sensor fusion approach combined with an automated 3-tier escalation workflow represents a significant advancement over single-sensor traditional solutions.

- **Practical applicability** The proposed system delivers a highly scalable and cost-effective solution ready for real-world deployment in nursing homes, assisted living facilities, and private smart homes. By automating continuous patient monitoring and generating a historical "Health Digital Twin", the system significantly reduces the physical and mental workload of medical staff and family members, minimizes false alarms, and ensures rapid medical intervention during critical domestic accidents.

14. Approval Section

Role	Name	Decision	Signature	Date
Faculty Mentor		☐ Approved ☐ Revise ☐ Reject		
Program Chair		☐ Approved ☐ Revise ☐ Reject		

CAPSTONE PROJECT PROPOSAL EVALUATION RUBRIC

Project title: Smart Elderly Care AI - A multi-modal AI system that actively monitors, forecasts, and cares for the elderly aged 60-89 years

TeamID: C1SE.34

Date:

◆ Evaluation Criteria

No.	Criteria	Weight	Key Focus	Score/mark
1	Problem Definition	15%	Clear, real-world relevance, stakeholders	
2	Complex Engineering Problem	20%	Meets ≥3 complexity characteristics	
3	Solution Feasibility	15%	Technical soundness, initial architecture	
4	Design Constraints	15%	Security, performance, cost, scalability ...	
5	Global & Ethical Impact	10%	Ethics, societal, environmental factors	
6	Innovation	10%	Novelty, originality	
7	Scope & Planning	10%	Feasibility within 14–16 weeks	
8	Team Structure	5%	Roles, capability balance	
9	Total score	100%		

Scale: 1 (Poor) – 5 (Excellent)

◆ Final Decision

Total Score	Decision
≥ 75	✅ Approved
60–74	⚠️ Revise & Resubmit
< 60	❌ Reject

◆ Comments

- Strengths:
- Weaknesses:
- Required Revisions:

Evaluator's Signature